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Opinion

Better artificial intelligence does not mean better models of biology

Drew Linsley[1,*], Pinyuan Feng[2], and Thomas Serre[1]

Deep neural networks (DNNs) once showed increasing alignment with primate perception as they improved on vision benchmarks, raising hopes that advances in artificial intelligence (AI) would naturally yield better models of biological vision. However, we present accumulating evidence that this alignment is now plateauing – and in some cases worsening – as DNNs scale to human or even superhuman accuracy. This divergence between artificial and biological perception may reflect the acquisition of visual strategies distinct from those of primates, and these findings challenge the view that advances in AI will naturally translate to progress in neuroscience. We argue that vision science must chart its own course, developing algorithms grounded in biological visual systems rather than optimizing for internet data.

Highlights

A surprising feature of early deep neural networks (DNNs) was that they evolved into better models of primate vision as their object-recognition performance improved.

Despite achieving human-level accuracy, today's scaled-up DNNs are worse than ever at predicting primate vision, likely because they rely on visual strategies that are misaligned with primates.

Vision science must chart an independent course, developing deep learning approaches tailored to biological vision rather than following artificial intelligence benchmarks.

The transformative promise of deep learning for vision science

One year after AlexNet – the **deep neural network (DNN)** (see Glossary) that sparked the modern AI revolution – Yamins, DiCarlo, and colleagues [1–3] published a pivotal finding: units in task-optimized DNNs trained for perceptual tasks, such as object recognition, responded to images in similar ways to real neurons in primate **inferotemporal cortex (ITC)**. This discovery transformed vision science and was soon joined by a chorus of similar findings in other domains of neuroscience (e.g., [4,5]). Suddenly, DNNs were not just systems for categorizing dogs and cats; they were now models of biology that could settle long-standing debates about neural computations and their links to perception. For example, the observation that DNNs pretrained for object classification better predicted IT responses provided computational backing to the theory of core object recognition – the idea that primate visual circuitry is optimal for rapid and invariant object recognition [6]. Perhaps most importantly, these newfound capabilities of DNNs implied that optimizing them for computer vision tasks and AI more broadly might uncover the principles that shape biological vision in the process [3,7]. Advances in AI through engineering could now lead the way and biological discovery would follow as a byproduct.

In the decade since AlexNet, DNNs have advanced dramatically, and current state-of-the-art models rival or surpass humans on nearly all vision benchmarks [8,9]. This progress has almost entirely been driven by an exponential scale-up of DNNs, a trend that was catalyzed by the introduction of **Transformers** [10] – attentional architectures that are purpose-built for the parallel processing capabilities of graphics processing units (GPUs) – and the observation that DNN performance on nearly all benchmarks improves predictably as the number of model parameters and amount of data used for training increases [11,12].

Nevertheless, today's state-of-the-art DNNs, including models like those from OpenAI, Anthropic, and Google – estimated to contain trillions of parameters trained on large proportions of the internet – still behave in strange ways; for example, stumbling on simple problems like

[1]Brown University, Providence, RI, USA
[2]Columbia University, New York, NY, USA

*Correspondence: drew_linsley@brown.edu (D. Linsley).

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counting or predicting viewpoints that seem trivial to humans, while excelling at a wide array of more complex ones [13–15]. This chasm between the human-level accuracy and alien-like behavior of today's DNNs raises a critical question for vision science: has the relentless engineering of DNNs to improve on AI benchmarks continued to bring concomitant improvements for modeling brains and behavior, or has it steered these systems away from biological principles altogether?

How deep learning has reshaped vision science

Task-optimized deep learning [16,17] has become extremely popular in computational neuroscience for modeling the brain for multiple reasons. It is the state-of-the-art modeling technique for system identification and predicting primate neural responses to images, an application which is fundamental for neuroprosthetic device development [18–21] and *in silico* studies of function [22–24]. Task-optimization can also be used to glean insights into the principles that sculpt visual circuitry.

For instance, early work found that DNNs trained for object classification predicted neural responses in **V4** and ITC better than models trained end-to-end for neural prediction [1,2]. This suggested that primate visual systems are organized around an objective to recognize objects in the world [4,25–39]. Notably, self-supervised learning approaches, which do not explicitly optimize for object classification, can lead to models that are nearly as effective at predicting ITC responses [27,28,40–43]. While task-optimization clearly matters for modeling primate vision, the relationship between specific tasks and neural organization remains an open question.

Task-optimization has also proven to be an effective strategy for modeling human perception. Most prominently, it has been repeatedly demonstrated that the alignment between human and DNN object-recognition decisions in psychophysics experiments increases with model accuracy on computer vision benchmarks [8,9,44–46]. This convergence is more than just a consequence of fewer mistakes; even the patterns of errors produced by these DNNs have come to resemble those of humans [9,47]. Task-optimized DNNs are capable of predicting human judgments across diverse tasks, including tests of local versus global processing biases [48], the perceived semantic similarity of object images [49,50], bottom-up image saliency [51,52], and the 3D properties of objects and scenes [13,53,54].

Others have found that co-training DNNs for object classification and predicting primate neural data can help models better predict human object classification psychophysics [55–57]. However, even with task-optimization, there are still multiple features of human perception that have proven challenging to model without introducing additional mechanisms into the model, such as perceptual phenomena like illusions [58–64] and metamers [65–68] – the case when two materially different stimuli are perceived as identical.

Inspired by the success of engineering benchmarks in advancing AI, the visual neuroscience community has developed its own benchmarks to systematically evaluate the prediction accuracy of models of biological vision [69,70]. A particularly influential initiative is **Brain-Score**, an online benchmark with a living leaderboard that allows anyone to submit and evaluate models on private data from dozens of neural and psychophysics experiments [7,71]. By tracking how DNNs perform on biological benchmarks as they improve on engineering ones, Brain-Score offers essential insights into how advances in AI have translated to biological vision.

The effectiveness of task-optimization has declined for modern DNNs

Despite its early transformative impact, optimizing DNNs for image tasks such as object classification or for learning from image datasets through **self-supervised training** has become an

Glossary

Attribution maps: visualizations that highlight regions of an input image that influence a neural network's predictions, providing insights into model decision-making processes and feature importance for specific classifications.
Brain-Score: an online benchmarking platform for systematically evaluating the accuracy of computational models at predicting neural and behavioral data.
Convolutional neural networks (CNN): DNNs models that apply multiple sets of learnable filters across all positions of an input, with each filter detecting patterns that are needed to solve the objective specified for the model. The same weights are shared across spatial locations. When layers of convolutional filters are stacked, deeper layers apply spatially local filtering to the outputs of previous layers. This process builds hierarchical representations, progressing from simple features to complex patterns.
Deep neural network (DNN): a computational model consisting of multiple processing layers that learn representations of data with multiple levels of abstraction, inspired by the neural networks in the brain. Convolutional neural networks and Transformer networks are special cases.
ImageNet: a large-scale dataset of labeled images spanning thousands of object categories, widely used as a benchmark for training and evaluating object-recognition systems.
Inferotemporal cortex (ITC): a region in the primate visual cortex responsible for high-level visual processing, particularly object recognition and categorization.
Neural alignment: the degree to which neural activity patterns in trained models correlate with or predict neural responses in biological systems when presented with the same stimuli.
Self-supervised training: training models with unlabeled data by creating supervisory signals from the data itself rather than human annotations. In computer vision, this typically involves solving pretext tasks, such as filling in masked image regions. Versions of self-supervised training are used in the training pipelines for large-scale generative models, like chatbots.
Transformer: a deep learning architecture that relies on self-attention mechanisms to process and relate different positions in input sequences,

increasingly unreliable strategy for modeling biological vision. We first observed this decline in the Brain-Score benchmark [32,72]. One of the experimental datasets included in Brain-Score is the same one used by Yamins, DiCarlo, and colleagues in their seminal work [73]. Brain-Score, however, updates those results by including the **neural alignment** of contemporary DNNs with the ITC neurons recorded in that dataset (Figure 1A, results replotted from Brain-Score.com). As newer DNNs have been added over recent years, a striking trend has emerged: DNNs continued to improve as models of ITC until they reached around 70% accuracy (chance is ~0.1%) on **ImageNet**, at which point they explained approximately half of the neural variance. Beyond this point, DNN neural alignment scores have declined even as their ImageNet accuracies have improved. In other words, optimizing DNNs for object classification has proven ineffective for building better models of ITC neurons in this dataset.

This trend of worsening neural alignment in image-classification optimized DNNs appears to be general. Indeed, we found a similar trend in recordings taken from a completely different study [32], in which monkeys were instructed to fixate at different locations across natural images as posterior ITC neurons were being recorded (Figure 1B). In this experiment, we benchmarked the neural alignment of a large, diverse, and highly controlled set of DNNs from the Pytorch Image Model library (TIMM) (https://github.com/rwightman/pytorch-image-models), which represented a decade's worth of progress in computer vision and included all of the popular architectures (**convolutional neural networks**, **Vision Transformers**, and hybrids of these two) and pretraining approaches (e.g., first training on ImageNet-21K [74], using self-supervised training approaches like DINO [75], or using multimodal learning approaches with image-language modeling approaches like CLIP [76]) over that period (note that Figure 1B depicts an updated version of the analysis in [32] with models since 2025).

Each of the DNNs was evaluated on two objectives: (i) accuracy on a multilabel ImageNet variant that was benchmarked against humans [77], and (ii) alignment with primate ITC neuron responses to specific regions of natural images [73] using the Brain-Score procedure [7,71]. What made this object classification benchmark unique was that it addressed a key issue in the original ImageNet, where multiple objects in some images had confounded earlier attempts to evaluate standard single-object classification accuracy. It also contained performance from five human participants on 80% of the ImageNet validation set, enabling direct human-to-machine comparisons for object recognition performance.

Many of the DNNs in this set had object classification performance that rivaled or surpassed humans: 30% (111) were human-level (scoring within the human confidence interval) and 1% (4) outperformed humans (the EVA class of models [78]; Figure 1B). At the same time, these human-level DNNs had also devolved into poor models of primate vision. For example, a superhuman DNN (EVA-02, released in 2023, only included in our re-analysis) was as aligned with ITC as one from 2019 that was far below human-level (the MobileNetV3 [79] from 2019). We also localized the inflection point at which the previously positive correlation between task performance and neural alignment turned negative at the TResNet – a GPU performance-optimized implementation of the original ResNet [80,81]. Follow-up analyses by us and others, comparing model performance across benchmarks such as the original ImageNet and more challenging variants, have shown similar reversals in neural alignment [72,82].

To understand why modern DNNs have diverged from biological vision, we analyzed how architectural and training factors influence neural alignment across hundreds of models. When we compared older, less-accurate models with newer, more accurate ones (less vs. more accurate than the 'TResNet' DNN inflection point in neural alignment), a striking pattern emerged. For older

initially developed for natural language processing but later adapted for computer vision tasks like image or video processing.
Vision Transformer (ViT): a type of transformer architecture adapted for computer vision tasks that processes images as sequences of patches rather than through traditional convolutional operations.
V4: an intermediate visual area in the ventral stream of the primate visual cortex that is thought to process complex visual features like shape and color, and which serves as a bridge between early visual areas and higher-level object-recognition regions.

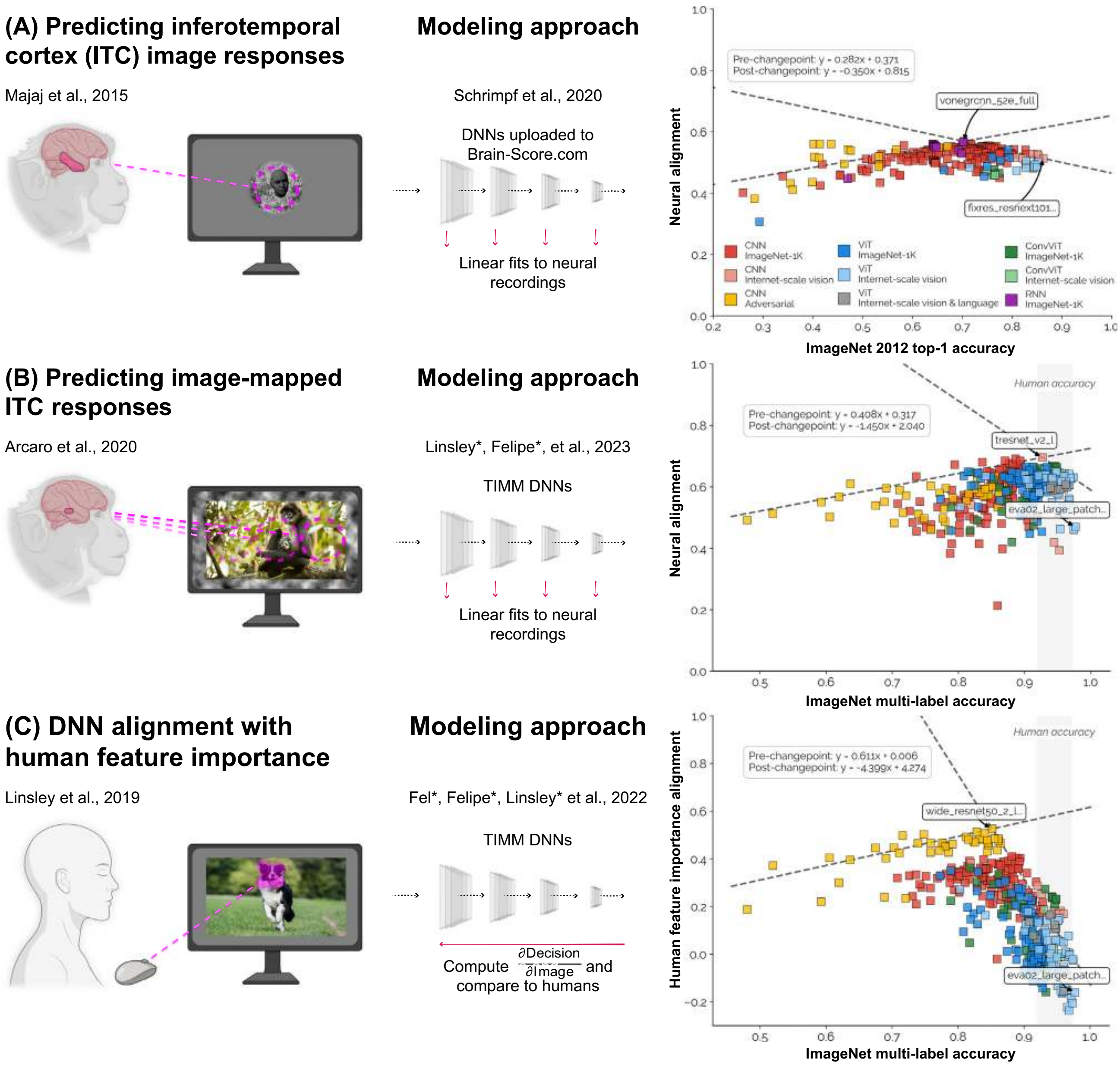

Figure 1. Deep neural networks (DNNs) increasingly diverge from biological vision as recognition accuracy improves. (A) DNNs optimized for visual tasks like object recognition are the standard for predicting image-evoked responses of neurons in the primate inferotemporal cortex (ITC) [1]. Units in each layer of an image task-optimized DNN are linearly fit to a training set of recordings and then the best-fitting layer is used to make predictions on independent recordings [7]. While there once was a strong and linear relationship between the object-recognition accuracy of a model and its ability to predict ITC responses to images, the sign of that correlation has flipped as models have surpassed 70% accuracy (chance is ~0.1%) on ImageNet. Each dot represents a different model. Broken lines denote trends of models on the pareto frontier of neural alignment (i.e., best-case biological predictions). The model representing the changepoint for the trends is the VOneGRCNN-52e-full [104], a recurrent convolutional neural network. (B) A similar pattern of results was found in another study of ITC [32,117], which sought to map the firing rates of neurons to different parts of the same image, yielding neural activity 'heatmaps' for images. The initially positive relationship between DNN object classification accuracy and neural predictivity flipped with the release of the 'TResNet' [88,113]. The gray patch denotes human accuracy on this version of ImageNet [74]. (C) One explanation for this flipped trend is that DNNs have learned to base their decisions on visual features that are different from those of humans (https://github.com/rwightman/pytorch-image-models) [88,114,115]. In this experiment [88], using data from [87], visual feature importance maps were derived from human participants who identified parts of images they considered diagnostic for recognition. DNN feature importance maps were then derived using standard

(Figure legend continued at the bottom of the next page.)

generations of models, better object-recognition and larger training datasets reliably improved neural alignment, confirming what the field had long believed [1,2,7,71] (Figure 2A). However, for today's high-performing models, this relationship has broken down entirely. Neither architectural choices (convolutional networks vs. transformers), model scale (parameters, layers, input size), nor training data (ImageNet alone vs. massive internet-scale datasets like ImageNet-21K or LAION-5B [83]) could explain which models better matched the properties of primate neurons. This disconnect is particularly striking given that these same factors – especially scale – are precisely what enabled DNNs to reach human-level performance. In other words, the computational strategies driving AI forward appear fundamentally different from those underlying biological vision.

Why does task-optimization now lead to worse models of biological vision?

One possible explanation for the declining effectiveness of task-optimized deep learning is that DNNs have begun to discover visual strategies that are challenging for biological visual systems to exploit as they have been scaled up. Early hints of this problem came from Ullman and colleagues [84], who found that early DNNs, such as AlexNet and VGG [85], reached their object-recognition decisions differently than humans. Humans exhibited an all-or-nothing reliance on diagnostic features, and the excision of a small number of pixels in an image was sufficient to render it no longer recognizable to a human observer. DNNs did not show the same tendency. These results revealed a new and surprising dimension for comparing human and DNN vision. At the same time, the study used a very small dataset and was limited in multiple ways [86], leaving open the question of whether these differences between humans and DNNs were mere experimental quirks or more fundamental.

To answer this question, we developed ClickMe, a scalable, gamified platform for collecting high-resolution feature importance maps for large natural image datasets [87–89]. In ClickMe, human participants highlight image regions that they find diagnostic for recognition, and while they do this, a DNN tries to recognize the object based only on dilated versions of those regions (Figure 1C). The faster participants can help the DNN recognize the object, the more likely they are to win a prize. The resulting maps were highly consistent across individuals and highlighted features significantly more diagnostic for human object classification than bottom-up saliency [87,88]. The diagnosticity of features in ClickMe maps was further validated in independent psychophysics experiments, showing that another group of participants was significantly better at classifying objects when using features highlighted in ClickMe than in bottom-up saliency maps. We collected these maps for a large proportion of ImageNet and compared their alignment with analogous feature-**attribution maps** from DNNs [90].

This experiment revealed a growing gap between artificial and human vision: as DNNs have advanced beyond human accuracy, the features they have learned for recognition have become progressively more distinct from those used by humans. For example, an 'adversarially-trained Wide ResNet50' achieved the highest alignment with humans, capturing ~50% of the variance in human feature importance maps. More accurate models showed precipitously declining human alignment, increasingly relying on features that differed from those used by humans – such as background textures, global statistics, and even watermark labels (Figure 3) [32,88,91]. Statistical analyses of DNN design factors confirmed this pattern (Figure 2B). While training data, architecture, and scale significantly predicted human feature alignment in older, less-accurate models (before the adversarially-trained Wide ResNet in Figure 1C), these

artificial intelligence interpretability methods based on the model's decision gradient. The alignment between humans and DNNs was measured by computing the correlation between these maps. The changepoint model is an 'Adversarially-trained Wide ResNet50' [116]. See [1,7,32,73,87,88,104,113–117]. Abbreviations: CNN, convolutional neural network; ViT, Vision Transformer.

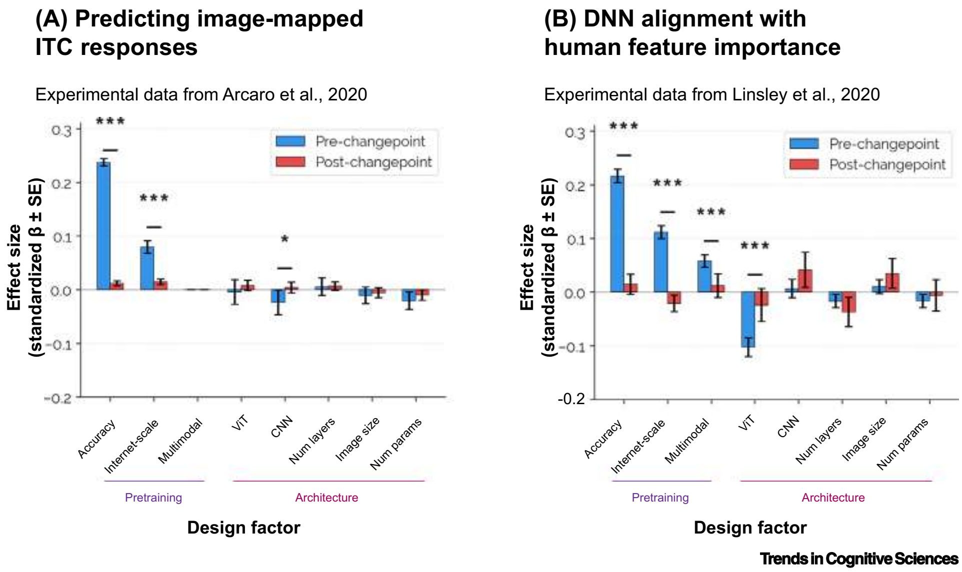

Figure 2. Deep neural network (DNN) design factors and biological alignment. We used analysis of variance (ANOVA) models to estimate how DNN design factors predict biological alignment on two experimental datasets: (A) predicting image-mapped inferotemporal cortex (ITC) responses [32,117], and (B) DNN alignment with human feature importance [87]. Our analysis estimated the design factor and alignment relationship for two groups of DNNs per experiment: those that fell before versus after the DNN 'change points' identified in Figure 1 and the main text, in which the positive relationship between ImageNet accuracy and biological alignment flipped to a negative one. The dependent variable in these ANOVAs was biological alignment and the independent variables described design factors for the models that produced these scores: accuracy on multilabel ImageNet (continuous), if the model was pretrained on an internet-scale image or multimodal dataset (binary factor: ImageNet-only versus ImageNet-21K/LAION-5B pretraining), if it was a convolutional neural network (CNN) or Vision Transformer (ViT) (binary), the number of layers (continuous), image resolution (continuous), and model parameter count (continuous). All independent variables were normalized to zero mean and unit variance across each experiment to make their effect sizes comparable. We fit separate ANOVAs for DNNs before versus after the change points in each experiment. We extracted parameter estimates for each independent variable and their robust standard errors from each ANOVA, which we refer to as 'Effect size' in the figure. In both experiments, the alignment of older and less accurate DNNs (before the change point) improved with internet-scale pretraining and better ImageNet accuracy. However, for newer, more accurate DNNs (after the change point), these relationships broke down entirely – neither architectural choices, model scale, nor training data could explain which models better matched biological vision. We used Wald tests to evaluate if the difference between parameter estimates for DNNs before versus after the DNN change points were significantly different: $t = (\beta\_pre - \beta\_post) / \sqrt{(SE\_pre^2 + SE\_post^2)}$. Significance markers (*, $P < 0.05$; ***, $P < 0.001$) denote differences between the two groups, highlighting how the computational strategies driving artificial intelligence progress have diverged from those underlying biological vision. See [32,87,117].

relationships disappeared entirely in more accurate models – mirroring the breakdown observed for neural predictivity (Figure 2A). These results underscore the fundamental shift in DNN optimization strategies that has taken place over recent years: the scale-up of DNNs to human-level performance has come at the cost of modeling biology. Today's state-of-the-art DNNs achieve their performance through a mixture of human-like and distinctly non-biological strategies. While these models retain some alignment with biological vision, they increasingly supplement evolutionarily-honed mechanisms with alternative – and often inscrutable – solutions that may be highly valuable for non-ethological tasks such as biomedical image analysis [92,93], but are less informative for understanding brains and behavior.

If task-optimized DNNs no longer produce better biological models, what will?

The magnificent promise of task-optimized deep learning was in its suggestion that all we needed was more engineering, including data and compute to precisely model biological vision. The now-

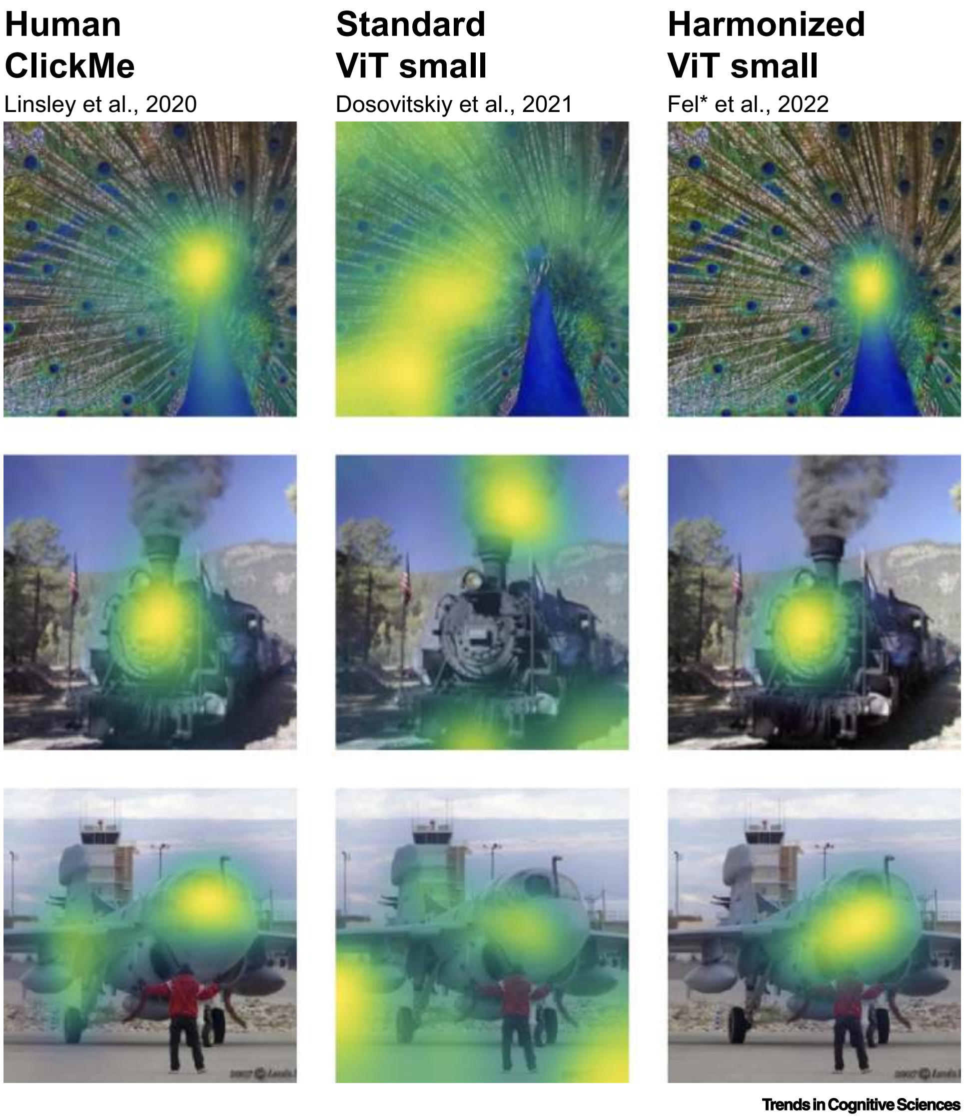

Figure 3. Deep neural networks (DNNs) can be trained to adopt human-like visual strategies. (Left) ClickMe [87] maps identify pixels humans believe are important for recognizing these object images. (Middle and right) Decision attribution maps for the same images from a Vision Transformer (ViT) [118] and harmonized ViT [88] highlight pixels each model uses for recognition. Without harmonization, ViTs often rely on contextual features like watermarks and backgrounds that are less important to humans. See [87,88,118].

waning effectiveness of this approach presents an opportunity for vision scientists to rethink how we model brains. The first question we must ask is whether DNNs are the right modeling framework to begin with [65].

We sought to answer this question by using a supervised learning approach, which we refer to as ‘harmonization’, that directly optimizes for biological alignment [88]. If DNNs are a reasonable modeling framework going forward, then using the harmonization method to guide them toward object decision attribution maps (Figure 1C) that are as close as possible to human feature importance maps (e.g., collected with ClickMe) during ImageNet training would validate that possibility. We found that harmonized DNNs not only relied on human-like visual features for object classification (Figure 3), but they also learned human-like sensitivity to adversarial attacks [91]. Furthermore, their image representations were significantly more aligned with ITC neurons than those of standard 'unharmonized' models [32].

While there have been recent calls for shifting the brain modeling paradigm away from deep learning [65], our harmonization experiments indicate that DNNs remain a viable approach for building predictive models of biological vision [94]. The problems we observe with DNNs trained on internet data, which are increasingly misaligned with biological vision (Figure 1), reflect a mismatch between the training objectives and the data. Co-training with biological data through harmonization can address this.

However, the breakthrough of task-optimized deep learning was that it could reveal principles and test mechanistic-level hypotheses through the principle that by optimizing for object recognition alone, we could naturally discover the computational strategies evolution selected for. Approaches like harmonization, by contrast, enforce biological alignment without providing insight into the origins of vision. Identifying these atomic components will require renewed focus on searching for data diets, behavioral objectives, and routines for training DNNs that guide them to perceive and encode the world like humans from the outset.

Rethinking task-optimization for biological discovery in the era of internet-scale compute

Computational models in vision science have long operated at smaller scales than frontier AI systems. This reflects at least three factors: the academic roots of vision science, a well-founded desire to rely on reductionism for testing mechanistic hypotheses, and the fact that most existing efforts to incorporate biological mechanisms into DNNs do not easily scale. For example, many biologically-inspired DNNs feature architectural constraints, such as recurrence [33,61,93,95–97] and different forms of feedback [59,60,98], which are known to play key roles in primate vision [99–102]. While these approaches are valuable for testing the impact of different mechanisms on complex behaviors [59], recurrent circuits are typically awkward fits for GPUs because the algorithms used to train them [96,100] make it difficult to scale to large datasets.

These scaling challenges are particularly problematic because biological alignment may emerge only from the right combination of data, objective, and circuit constraints when models are trained at very large scales. Moreover, it is possible that some of the computational strategies of biologically-inspired DNNs could be rediscovered by an easy-to-scale DNN trained with the 'right' data and objective [101,102], raising questions about the value of inducing these constraints in the first place. Thus, it may be that the most effective approach to reverse-engineering vision with DNNs is to bypass architecture questions altogether: begin with the same unbiased and scalable DNN architectures that are popular in computer vision (e.g., Vision Transformers), and focus efforts on searching for the data diets and training routines that cause this system to exhibit similar representations and behavior as biological vision. Once we identify data and training routines that successfully align task-optimized models with biology, we can then build a comprehensive explanation of biology [3,103] by incorporating other biological features that may require innate constraints or physical embodiment to emerge, such as topography [19–21] or neuronal noise [104].

We propose that better models of primate vision will emerge from training DNNs with biologically-grounded data diets and learning principles (see Outstanding questions). While others have made similar calls before [3,105–107], the field has yet to seriously pursue this direction at scales that may be infeasible for academic research, yet also necessary to observe biological features emerge from models. The exponential growth in computational resources over recent years now makes it feasible to move beyond the convenient but biologically implausible paradigm of training on large databases of static images [108,109]. Instead, we can explore how biological alignment emerges as models learn from temporally structured and embodied experiences – the same kinds of sensory streams and environmental pressures that actually shape developing visual systems.

Modern simulation environments and video generation techniques [110–112] have reached sufficient fidelity to create naturalistic training data on demand at scale, while advances in self-supervised training have sketched out spaces of behavioral objectives that can be systematically evaluated with these data. The key insight is that we should not be searching for a single 'correct' objective, but rather mapping the manifold of developmental and ecological constraints that naturally produce human-like vision. Only through this systematic exploration can we build models that predict biological vision and reveal its underlying principles, ultimately transforming our understanding of why we see the world the way we do in the process.

Concluding remarks

DNNs once demonstrated increasing alignment with primate vision as they improved on object-recognition benchmarks, suggesting that advances in artificial vision would naturally yield better models of biological vision. However, this alignment has plateaued – and in some cases worsened – as DNNs have scaled to human-level accuracy, likely because modern systems rely on visual strategies that diverge from those of biological systems. These findings underscore the need for vision science to chart its own course in computational modeling that is grounded in biological principles rather than purely optimized for internet data.

Outstanding questions

How can we create more biologically realistic training regimens for DNNs? Today's DNNs are trained on massive image collections that bear little resemblance to the temporally structured and rich, multimodal experiences that actually shape biological visual systems. How can we bridge this gap? Training DNNs within ecologically valid and dynamic environments using state-of-the-art computer graphics methods.

What learning principles will align DNNs with biological vision? The objective functions used in today's DNNs are designed with static image data in mind. What happens when we move our models to dynamic and embodied data collection? It is essential for the field to systematically explore objectives and data curricula that might cause DNNs to learn more human-like visual representations.

What role will biological constraints play in the search for data diets and learning principles? It is challenging to imagine how certain features of the brain – such as topographical organization and neural noise – could emerge without being explicitly induced. Determining which constraints are fundamental principles and which are byproducts of biological implementation will be crucial for developing models that accurately predict both neural responses and behavior.

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Declaration of interests

The authors declare no competing interests.

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